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EDUCATION

Carnegie Mellon University | *Master of Science in Machine Learning* | GPA: 4.0/4

December 2026

- *Research*: Sharpness-Aware Pretraining Mitigates Catastrophic Forgetting [[ICML'26](#)] with **Prof. Aditi Raghunathan**
- *Coursework*: Deep RL, ML-Systems, Convex Optimization, Probabilistic Graphical Models, Advanced Introduction to ML
- *Teaching Assistant*: AI for Scientific Computing with **Prof. Andrej Risteski**

Indian Institute of Technology, Delhi | *Bachelor of Technology in Engineering Physics* | GPA: 9.0/10

May 2023

WORK EXPERIENCE

HippocraticAI | Research Scientist Intern | *Proactive Personalization for Healthcare LLMs*

May 2026 - Aug 2026

- Conducted empirical research on **memory-aware healthcare conversational agents**, analyzing 70K+ real-world voice conversations to derive a data-driven **taxonomy of memory usage** for proactive personalization.
- Designed a **human-aligned evaluation framework** for **proactive personalization** by decomposing the task into memory retrieval (*when to personalize*) and memory integration (*how to personalize*), with pairwise preference rubrics evaluating naturalness and personalization quality (coherence, engagement) to enable **DPO-based alignment**.

Google DeepMind | Research Fellow | *Modular LLMs for Continual Learning*

July 2024 - July 2025

- Developed a **modular multi-model** architecture with dynamic routing over **64 expert models**, fusing the selected expert with a generalist via **cross-attention** and outperforming a monolithic model 2× its size.
- Evaluated **generalist-specialist model fusion** for continual domain adaptation and **compositional tasks** (e.g., coding + multilingualism), demonstrating gains over LoRA across varied token and parameter budgets. [[Gemini-2.5 technical report](#)]
- Engineered scalable **JAX-based** training, evaluation & inference pipelines for multi-model architectures in **Gemini** codebase.

Microsoft Research | Research Associate | *Multilingual Evaluation of LLMs*

May 2023 - June 2024

- Built a culturally-nuanced **LLM evaluation framework** across **10 Indic languages**, conducting **120K+ human and LLM judgments** across 30+ models to study multilingual and cross-cultural evaluation. [[publication: EMNLP'24](#)]
- Developed a **38-language toxicity benchmark** spanning 8 harm categories and evaluated 10 LLMs, revealing failures in detecting microaggressions; studied **QLoRA finetuning** across 90+ models, analyzing cross-lingual transfer and catastrophic forgetting; benchmarked 20 LLMs across **83 languages** and 23 tasks. [[publications: AAAI'25, ACL Findings'24, NAACL'24](#)]
- Developed and deployed **Shiksha Copilot**, a GenAI tool for creating educational content, across **100+ schools** in India.

ACADEMIC PROJECTS

Sharpness-Aware Pretraining Mitigates Catastrophic Forgetting | *Prof. Aditi Raghunathan*

Sept 2025 - April 2026

- Proposed **sharpness-aware pretraining**, reducing catastrophic forgetting by up to **80%** across **20M–150M** models after supervised fine-tuning on 5 datasets and under 4-bit quantization; validated findings at **1B** scale. [[publication: ICML'26](#)]
- Conducted **3,500+** experiments & analyzed Hessian curvature to connect **loss landscape sharpness** with continual adaptation; characterized effects of **SAM**, larger learning rates, & shorter annealing schedules on post-training performance.

Adaptive Expert Placement for Efficient MoE Inference | *15-642 ML Systems, CMU*

March 2026 - May 2026

- Built a profiling pipeline to analyze **expert routing, compute, and communication costs** in MoE LLMs, using SPEED-Bench workloads and profiling **vLLM MoE GroupGEMM kernels and all-to-all communication**.
- Developed a **routing-aware expert placement** method that optimizes **expert-to-GPU assignments** for each layer, achieving up to **17% predicted speedup** over standard blocked and interleaved placements by reducing **load imbalance** across GPUs.

SELECTED PUBLICATIONS

Sharpness-Aware Pretraining Mitigates Catastrophic Forgetting. [[Spotlight @ICBINB Workshop ICLR'26](#)][[ICML'26](#)]

RTP-LX: Can LLM Evaluate Toxicity in Multilingual Scenario? [[AAAI'25: AI for Social Impact track](#)]

PARIKSHA: Investigation of Human-LLM Evaluator Agreement on Multilingual & Multi-Cultural Data. [[EMNLP'24](#)]

TECHNICAL SKILLS

ML/DL: Pretraining, Continual Learning, Agent Memory, Large Language Models (LLMs), Evaluations, Reinforcement Learning
Deep Learning, Multilingualism, Natural Language Processing (NLP), Conversational AI, ML System **Tools**: Git, Docker, CUDA
Frameworks/Libraries: PyTorch, JAX, LangChain, NumPy, Pandas, Matplotlib, Scikit-learn **Languages**: Python, SQL, C/C++